PROJECT REVIEW – 1

Explainable Real Estate Analytics: A Comparative Analysis of Machine Learning Frameworks for House Price Prediction

Project Domain: DATA ANALYTICS & VISUALIZATION

Department: Computer Science and Engineering

University: KLH UNIVERSITY

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Academic Year: 2026–2027

1. REFINED TITLE

Explainable Real Estate Analytics: A Comparative Analysis of Machine Learning Frameworks for House Price Prediction

The title is refined from the initial comparative-analysis topic based on recent research in machine learning-based property valuation. The proposed work retains comparative evaluation and extends it by integrating feature engineering, ensemble learning and Explainable Artificial Intelligence (XAI).

2. PROBLEM STATEMENT

House prices are influenced by several factors such as location, property area, number of rooms, property age, quality and amenities, making accurate prediction challenging. Different machine learning models capture these relationships differently, while complex ensemble models may be difficult to interpret. Existing studies have explored machine learning, ensemble learning and Explainable Artificial Intelligence for property valuation, but there is still scope for an integrated and practical framework. Therefore, this project aims to develop an explainable ensemble learning framework that improves house price prediction while identifying the major factors influencing the predicted price.

3. LITERATURE SURVEY

The following recent journal papers were reviewed to understand current machine learning approaches, ensemble methods, explainability techniques and research limitations in house-price prediction and property valuation.

Year	Author(s)	Paper	Method / Finding	Gap / Relevance
2026	Mammadzade, M.	Machine Learning Algorithms for House Price Prediction: A Comparative Study	Compares Linear Regression, Ridge Regression, Decision Tree and Random Forest for house-price prediction using the Ames Housing dataset.	Provides useful model comparison; scope remains for an integrated ensemble and XAI framework.
2025	Cao, Y., Xu, Y., Ji, Y., & Wang, H.	Automated Machine Learning for Residential Property Valuation	Proposes an AutoML framework for residential property valuation with automated model selection and optimization.	Strong automation, but relatively complex and not primarily focused on transparent prediction.
2025	Deng, L., & Zhang, X.	Boosting the Accuracy of Property Valuation with Ensemble Learning and Explainable Artificial Intelligence: The Case of Hong Kong	Uses ensemble learning for property valuation and XAI to identify important price determinants.	Highly relevant, but based on Hong Kong property data and a domain-specific valuation setting.
2025	Maselli, G., & Nesticò, A.	Machine Learning Algorithms and Explainable Artificial Intelligence for Property Valuation	Studies machine learning models for property valuation and applies SHAP and feature-importance techniques.	Shows the value of XAI; scope remains for combining comparative evaluation and stacking in one framework.
2025	Viktoratos, I., & Tsadiras, A.	Advancing Real-Estate Forecasting: A Novel Approach Using Kolmogorov–Arnold Networks	Investigates Kolmogorov–Arnold Networks for real-estate forecasting and compares them with other approaches.	Advanced model is more complex and does not primarily address interpretable ensemble prediction.
2025	Adonia, N., Mwangi, W. R., & Michael, K. W.	Optimizing Multivariate Adaptive Regression Splines with Coordinate Descent to Accurately Select the Best Model for House Price Prediction	Uses optimized Multivariate Adaptive Regression Splines with coordinate descent for house-price prediction and model selection.	Focuses on one optimized regression approach rather than complementary ensemble models with XAI.
2025	Karanikolas, N., Kyriakidou, E., & Athanasouli, E.	Artificial Intelligence and Real Estate Valuation: The Design and Implementation of a Multimodal Model	Develops a multimodal AI approach for real-estate valuation using different property information sources.	Multimodal modelling can require additional data and computational resources.
2025	Wang, X., & Kee, T.	Integrating Shapley Value and Least Core Attribution for Robust Explainable AI in Rent Prediction	Investigates Shapley-based attribution methods for explaining property rent predictions.	Strong focus on explainability, but the task is rent prediction rather than house-sale-price ensemble modelling.
2024	Jafary, P., Shojaei, D., Rajabifard, A., & Ngo, T.	Automating Property Valuation at the Macro Scale of Suburban Level: A Multi-Step Method Based on Spatial Imputation Techniques, Machine Learning and Deep Learning	Combines spatial imputation, machine learning and deep learning for macro-scale property valuation.	Powerful but spatially and computationally complex for a compact structured-data project.
2026	Kmen, C., Navratil, G., Kattenbeck, M., & Giannopoulos, G.	Evaluating Human–Machine Collaboration Through a Comparative Analysis of Experts, Machine Learning, and Hybrid Approaches in Real Estate Valuation	Compares machine learning, expert valuation and hybrid human–machine approaches using real-estate transaction data.	Focuses on human–machine collaboration rather than a standalone explainable ensemble framework.

Literature synthesis: Recent studies indicate that ensemble learning, automated machine learning, advanced neural models and Explainable AI are increasingly used in property valuation. Tree-based models and ensemble methods remain effective for structured property data, while SHAP and other attribution methods improve interpretability. However, advanced spatial, multimodal and AutoML frameworks may require more data and computational resources. This creates scope for a practical, reproducible framework combining comparative model evaluation, ensemble learning and explainability.

4. RESEARCH GAPS IDENTIFICATION

- Many studies compare individual machine learning models, but there is scope for integrating complementary high-performing models into a single ensemble framework.
- Although recent work combines ensemble learning and Explainable AI, several approaches are region-specific, spatially intensive or designed for specialized valuation environments.
- Advanced AutoML, multimodal and deep-learning approaches can improve prediction but increase implementation and computational complexity.
- Prediction accuracy is often emphasized more than the interpretability of individual predictions.
- A practical framework combining feature engineering, model comparison, stacking and SHAP-based explanation remains a suitable research direction for structured house-price data.

5. OBJECTIVES

1. To preprocess and prepare house-price data using missing-value treatment, categorical encoding, outlier handling and feature engineering techniques.
2. To implement and compare Linear Regression, Random Forest, XGBoost and CatBoost using K-Fold Cross-Validation and evaluation metrics such as MAE, RMSE and R^2.
3. To develop a Stacking Regressor by combining the best-performing machine learning models to improve house-price prediction performance.
4. To apply SHAP-based Explainable Artificial Intelligence to identify and interpret the most influential features contributing to house-price predictions.

6. NOVELTY / INNOVATION

The novelty of the proposed work is not the individual use of ensemble learning or SHAP, since these techniques already exist in recent literature. The proposed innovation is their integration into a single practical house-price prediction pipeline.

- Systematic feature engineering and preprocessing before model comparison.
- Comparative evaluation of multiple regression models followed by selection of complementary high-performing models.
- Use of a Stacking Regressor to combine selected models instead of relying only on one individual predictor.
- Application of SHAP to the final model to provide understandable feature-level explanations for house-price predictions.

7. FEASIBILITY

1. Technical Feasibility

- Implemented using **Python**.
- Uses open-source libraries such as **Scikit-learn, XGBoost, CatBoost, and SHAP**.
- Public housing datasets are available for experimentation.
- Can be executed on a standard computer.

2. Economic Feasibility

- No specialized paid software is required.

- Uses freely available open-source tools and libraries.
- Low overall development and implementation cost.

3. Operational Feasibility

- Suitable for an academic machine learning project.
- Workflow is simple and manageable.
- Can be completed within the planned project duration.
- Requires only standard computing resources.

8. PROJECT PLAN

Week	Planned Activity
Phase 1	Literature survey; finalize title, problem statement and research gaps; collect, clean and preprocess the dataset.
Phase 2	Feature engineering and exploratory data analysis; prepare relevant features for model development.
Phase 3	Implement individual machine learning models; perform cross-validation, model comparison and hyperparameter tuning.
Phase 4	Develop and evaluate the Stacking Regressor; compare ensemble performance with individual models; perform SHAP-based explainability and feature analysis.
Phase 5	Testing and final evaluation; analyze results; prepare documentation, report and presentation.

9. TEAM DETAILS

S.No.	Team Member	Roll No.	Responsibility
1	Shalini . Ch	2420030284	Literature Survey, Data Preprocessing & Feature Engineering
2	M.J.U. Harika	2420030453	ML Model Development, Model Comparison & Ensemble Learning
3	K . Aneesha	2420030589	SHAP Analysis, Result Analysis, Documentation & Presentation

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